Reanalysis and secondary findings
A published case: a negative trio exome, re-read at age five, gives the answer

Summary for doctors and healthcare teams, written in our own words from a published, peer-reviewed case report. Please read the original article (link below).

Source: Hochman L, Drummond A, Morgan K. *Cerebellar Ataxia, Impaired Intellectual Development, and Disequilibrium Syndrome-2: A Case Report.* Cureus 2025;17(1):e78066. doi:10.7759/cureus.78066. Florida State University College of Medicine and University of South Florida Health, USA. Open access (CC BY 4.0). doi.org/10.7759/cureus.78066

The patient

- Girl, twin A of a dizygotic pregnancy; term birth; non-consanguineous family; negative family history
- Neonatal hypoglycaemia and apnoeic episodes; early MRI and EEG unremarkable
- Developmental concerns, feeding difficulty and unsteady gait in infancy
- Cerebellar ataxia diagnosed at 18 months
- MRI: vermian-predominant cerebellar atrophy

Testing and reanalysis

- Initial work-up nondiagnostic: **trio exome with mitochondrial DNA analysis**, chromosomal microarray, ataxia gene panel, SMA testing, metabolic studies
- At age 5, after the family moved and she was re-referred to genetics, **the same trio exome data were reanalysed**
- Result: **homozygous WDR81 c.2567C>T, p.(P856L)**; each parent heterozygous
- Diagnosis: **CAMRQ2** (cerebellar ataxia, impaired intellectual development and disequilibrium syndrome 2), autosomal recessive

What it meant

No new sample or new sequencing was needed: the answer came from **re-reading existing data**. Her course is milder than most published cases: she walks independently, has age-appropriate speech and attends school at grade level with tutoring. She continues physiotherapy and occupational therapy. The authors suggest early diagnosis and therapy may have helped (a proposal, not a proven effect) and highlight the importance of **periodic re-evaluation of genetic data**. With both parents carriers, the family can be counselled about recurrence.

Talking points for the doctor visit

- A negative exome is not the last word; ask about reanalysis, especially with new symptoms or after time has passed.
- Re-evaluate any VUS first; send the lab updated clinical details.
- Before testing, record whether the patient wants secondary findings reported.
- Prenatal testing in India: written consent under the PCPNDT Act; disease testing only; fetal sex never disclosed.

EPISODE 10 · QUICK GUIDE

Reanalysis and secondary findings

Topic	Key points
What reanalysis is	Re-reading existing exome or genome data with current knowledge: new gene-disease links, updated databases and better software. Usually no new sample.
Where new diagnoses come from	Radboud cohort (31 new diagnoses): new or broadened gene-disease associations 42%; resequencing 29%; improved pipelines 19%; segregation or functional follow-up 10%.
Evidence	150 children with neurological disease: diagnostic yield rose from 31% to 53% over about 5 years (Schobers et al. 2022). Solve-RD, 6,004 families: 8.4% newly diagnosed by systematic reanalysis plus 4.1% by expert review, 12.6% overall (Laurie et al. 2025).
When to ask	New symptoms or signs; another affected family member; a VUS in the report (re-evaluate it first); time since the first test (the Radboud authors suggest about yearly re-evaluation). Send updated clinical details; ask the lab's policy and cost; keep old reports and data.
Secondary findings	Medically actionable results unrelated to the test question, e.g. hereditary cancer risk, some heart conditions, familial hypercholesterolaemia. ACMG SF v3.3 (2025) lists 84 genes; ABCD1, CYP27A1 and PLN were added.
Consent	Before testing, record whether the patient wants secondary findings; discuss children carefully; results may matter for relatives.

References

- Hochman L, Drummond A, Morgan K. Cureus 2025;17(1):e78066. doi.org/10.7759/cureus.78066
- Schobers G et al. Reanalysis of exome negative patients with rare disease: a pragmatic workflow for diagnostic applications. Genome Med 2022;14:66. doi:10.1186/s13073-022-01069-z
- Laurie S et al. Genomic reanalysis of a pan-European rare-disease resource yields new diagnoses. Nat Med 2025;31:478-489. doi:10.1038/s41591-024-03420-w
- ACMG SF v3.3 list for reporting of secondary findings in clinical exome and genome sequencing. Genet Med 2025. doi:10.1016/j.gim.2025.101454

Watch the episode: healthtestsimplified.netlify.app/#g010 · Facebook & YouTube: Health Test Simplified